Solution to Pun-ny Math

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1 Key/Code

A = 0	F = 5	K = 10	P = 15	U = 20	Z = 25
B = 1	G = 6	L = 11	Q = 16	V = 21	Note: All the problems are given in ascending order of the resulting numbers
C = 2	H = 7	M = 12	R = 17	W = 22	
D = 3	I = 8	N = 13	S = 18	X = 23	
E = 4	J = 9	O = 14	T = 19	Y = 24	

Although this solution outlines one of the ways you could have solved these problems, there are many other paths you could have taken to arrive at the same answer.

2 Answers

1. Question: What do you get when you multiply a famous place in New York City by itself?
   - Answer: **TIMES SQUARE**
     (a) (Part i's answer) - [(Part b's answer) + (Part e's answer)] $= 20 - (4 + 16) = \mathbf{0}$
     (b) (Letter used twice): The only number whose principal square root is half the number itself $= \mathbf{4}$
     (c) $2^4 - 2^3 = 16 - 8 = \mathbf{8}$
     (d) (f's answer)$-5 = 17 - 5 = \mathbf{12}$
     (e) $(5! - 4!) \div 6 = [(5 \times 4 \times 3 \times 2 \times 1) - (4 \times 3 \times 2 \times 1)] \div 6 = (120 - 16) \div 6 = 96 \div 6 = \mathbf{16}$
     (f) $\sqrt{300 - 11} = \sqrt{289} = \mathbf{17}$
     (g) (Letter used twice): $6 + 324 \div 6 - 42 = 6 + 54 - 42 = 60 - 42 = \mathbf{18}$
     (h) Find the positive difference between 2116 and the year this year's entering freshman class (2021-2022) will graduate high school. Now flip the resulting number's digits (for example, the number 48 would result in 84) $= 2116 - 2025 = 91$ which becomes **19**
     (i) $(2021 \div 47) - (2020 \div 202) - (2019 \div 673) - (2018 \div 1009)$
     $- (2017 \div 2017) - (2016 \div 288) = 43 - 10 - 3 - 2 - 1 - 7 = 43 - 23 = \mathbf{20}$
   - Letters: A, E, E, I, M, Q, R, S, S, T, U. Unscramble and you should get **TIMES SQUARE**
2. Question: What do you call brothers that love math?
   - Answer: **ALGEBROS**
     (a) (g's answer)-[(d's answer)+(e's answer)] $= 17 - (6 + 11) = \mathbf{0}$
     (b) $|1^2 - 2^2| \div 3 = |1 - 4| \div 3 = 3 \div 3 = \mathbf{1}$
     (c) $(6 - 5) \times 4 = 1 \times 4 = \mathbf{4}$
     (d) $7 + 8 - 9 = 15 - 9 = \mathbf{6}$
     (e) $10 - 11 + 12 = 12 - 1 = \mathbf{11}$
     (f) $13 + |14 - 15| = 13 + 1 = \mathbf{14}$

   (g) $16 - 17 + 18 = 18 - 1 = \mathbf{17}$
   (h) $19 + 20 - 21 = 39 - 21 = \mathbf{18}$

- Letters: A, B, E, G, L, O, R, S. Unscramble and you should get **ALGEBROS**

3. Question: What did the triangle say to the circle?

- Answer: **YOU'RE POINTLESS**
   (a) (Letter used twice): (f's answer)-(c's answer) $= 20 - 18 + (13 - 11) = 2 - 2 = \mathbf{0}$
   (b) $1 + 2^2 + 3 = 1 + 4 + 3 = \mathbf{8}$
   (c) First Digit $= |4 - 5|$, Second Digit $= |6 - 7|$, **11**
   (d) $8 - 9 + 10 - 11 + 12 - 13 + 14 - 15 + 16 - 17 + 18 = 8 + 1 + 1 + 1 + 1 + 1 = \mathbf{13}$
   (e) (Letter used twice): $18 - ||19^2 - 20^2| - |21^2 - 22^2|| = 18 - |361 - 400| - |441 - 484|| = 18 - |39 - 43| = 18 - 4 = \mathbf{14}$
   (f) $|23 - 24| \times (25 + 26)$, flip the resulting number's digits $= 1 \times 51 = 51$ which becomes **15**
   (g) $(2930 - 2728 - 32) \div 10 = (202 - 32) \div 10 = 170 \div 10 = \mathbf{17}$
   (h) (Letter used twice): $31 + 33 - 35 - 11 = 64 - 35 - 11 = 29 - 11 = \mathbf{18}$
   (i) $38 \div (36 - 34) = 38 \div 2 = \mathbf{19}$
   (j) $40 \div (44 - 42) = 40 \div 2 = \mathbf{20}$
   (k) $|42 - 46|! = 4! = 4 \times 3 \times 2 \times 1 = \mathbf{24}$
- Letters: Letters: E, E, I, L, N, O, O, P, R, S, S, T, U, Y. Unscramble and you should get **YOU'RE POINTLESS**

4. Question: What do you call a political party in favor of agriculture?

- Answer: **PRO-TRACTORS**
   (a) (g's answer)-[(b's answer)+(e's answer)] $= 19 - (2 + 17) = \mathbf{0}$
   (b) This number is the only even prime number $= \mathbf{2}$
   (c) (Letter used twice): This number has a prime factorization of (b's answer)(7) $= 2 \times 7 = \mathbf{14}$
   (d) This number is the smallest two digit positive multiple of 5 that has only odd factors $= \mathbf{15}$
      – 15: 1, 3, 5, 15 (all numbers are odd)
   (e) (Letter used twice): This is the second largest prime under 20 $= \mathbf{17}$
   (f) This number has a prime factorization of (b's answer)(e's answer-c's answer)$^2 = 2 \times (17 - 14)^2 = 2 \times 3^2 = 2 \times 9 = \mathbf{18}$
   (g) (Letter used twice) This is the largest prime under 20 $= \mathbf{19}$
- Letters: A, C, O, O, P, R, R, S, T, T. Unscramble and you should get **PRO-TRACTORS**

5. Question: What do you call a male mathematician who spent all summer on the beach?

- Answer: **A TAN GENT**
   (a) (Letter used twice): (e's answer)-[(c's answer)+(d's answer)] $= 19 - (6 + 13) = \mathbf{0}$
   (b) The function of p$\heartsuit$q is defined as 2p + 3q. If x$\heartsuit$36 = 188, find x-36 $= 40 - 36 = \mathbf{4}$
   (c) The function of h❖k is defined as $h^2 - k^2$. If x❖21 = 184, find $|x| - 19 = 25 - 19 = \mathbf{6}$
   (d) (Letter used twice): The function of t☆u is defined as $-3(t + u)$. If 5☆x = 9, find 5 - x $= 5 - (-8) = \mathbf{13}$
   (e) (Letter used twice): The function of y$\sqcap$z is defined as $2(z - y)^2$. If x$\sqcap$20 = 722 and $y < z$, find $20 - x = 20 - 1 = \mathbf{19}$
- Letters: A, A , E, G, N, N, T, T. Unscramble and you should get **A TAN GENT**